Tutorial 3'

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Let $f(x) = -x^3 - \cos x$. With $p_0 = -1$ and $p_1 = 0$, find p_3.

a. Use the Secant method. **b.** Use the method of False Position.

First notice that

$f(-1) = 0.4596$, $f(0) = -1$

both methods have the form

$$p_n = p_{n-1} - \frac{f(p_{n-1})\,[\,p_{n-1} - p_{n-2}\,]}{f(p_{n-1}) - f(p_{n-2})}$$

however, the false position choose to always bracket the solution.

a) $p_0 = -1$ $f(-1) = 0.4596$

$p_1 = 0$ $f(0) = -1$

$$p_2 = 0 - \frac{(-1)\,[\,0 - (-1)\,]}{(-1) - 0.4596} = -0.68512$$

$f(p_2) = -0.45276$

$$p_3 = p_2 - \frac{(-0.45276)\,[\,-0.68512 - 0\,]}{-0.45276 - (-1)}$$

$$= -1.25196$$

b) $p_2 = -0.68512$, $f(p_2) = -0.45276$

Since we always want to bracket the solution p_2 replaces p_1 not p_0.

$$p_3 = g(p_2, p_0)$$

$$= p_2 - \frac{f(p_2)\,[\,p_2 - p_0\,]}{f(p_2) - f(p_0)} = -0.84138$$

Use Newton's method with $\mathbf{x}^{(0)} = \mathbf{0}$ to compute $\mathbf{x}^{(2)}$ for each of the following nonlinear systems.

a. $4x_1^2 - 20x_1 + \frac{1}{4}x_2^2 + 8 = 0,$

$\frac{1}{2}x_1x_2^2 + 2x_1 - 5x_2 + 8 = 0.$

$$G(\underline{x}) = \underline{x} - J^{-1}(\underline{x})\,F(\underline{x})$$

such that

$$G(\underline{p}) = \underline{p} \quad \text{where} \quad F(\underline{p}) = 0$$

let us first find $J(x)$

$$J(x) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 8x_1 - 20 & \frac{1}{2}x_2 \\ \frac{1}{2}x_2^2 + 2 & x_1x_2 - 5 \end{bmatrix}$$

Thus, $x^{(k)} = G(x^{(k-1)})$

$= x^{(k-1)} - J^{-1}(x^{(k-1)})\, F(x^{(k-1)})$

$x^0 = (0,0)$ $\quad J(x^0) = \begin{bmatrix} -20 & 0 \\ 2 & -5 \end{bmatrix}$

$J^{-1}(x^0) = \frac{1}{100}\begin{bmatrix} -5 & 0 \\ -2 & -20 \end{bmatrix}$

$(x^1) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \frac{1}{100}\begin{bmatrix} -5 & 0 \\ -2 & -20 \end{bmatrix}\begin{bmatrix} 8 \\ 8 \end{bmatrix}$

$= \begin{bmatrix} 2/5 \\ 1.76 \end{bmatrix}$

$J(x^1) = \begin{bmatrix} -16.8 & 0.88 \\ 3.5488 & -4.296 \end{bmatrix}$

$J^{-1}(x^1) = \begin{bmatrix} -0.06222 & -0.01274 \\ -0.051395 & -0.2433 \end{bmatrix}$

$F(x^1) = \begin{bmatrix} 1.4144 \\ 0.61952 \end{bmatrix}$

$J^{-1}(x^1)\, F(x^1) = \begin{bmatrix} -0.095897 \\ -0.22342 \end{bmatrix}$

$x^2 = \begin{bmatrix} 2/5 \\ 1.76 \end{bmatrix} + \begin{bmatrix} 0.09589 \\ -0.22342 \end{bmatrix} = \begin{bmatrix} 0.49589 \\ 1.53658 \end{bmatrix}$

Find the first two iterations of the Jacobi method for the following linear systems, using $\mathbf{x}^{(0)} = \mathbf{0}$:

$$\begin{aligned} 3x_1 - x_2 + x_3 &= 1, \\ 3x_1 + 6x_2 + 2x_3 &= 0, \\ 3x_1 + 3x_2 + 7x_3 &= 4. \end{aligned}$$

Jacobi :

$x_1 = \frac{1}{3}\left[1 + x_2 - x_3\right]$

$x_2 = \frac{1}{6}\left[-2x_3 - 3x_1\right]$

$x_3 = \frac{1}{7}\left[4 - 3x_2 - 3x_1\right]$

$x^0 = (0,0,0)^t$

k	$x_1^{(k)}$	$x_2^{(k)}$	$x_3^{(k)}$
0	0	0	0
1	1/3	0	4/7
2	1/7	-5/14	3/7

0	0	0	0
1	1/3	0	4/7
2	1/7	-5/49	3/7

Gauss - seidel

k	$x_1^{(k)}$	$x_2^{(k)}$	$x_3^{(k)}$
0	0	0	0
1	1/3	-1/6	1/2
2	1/9	-2/9	13/6

Compute the condition numbers of the following matrices relative to $\|\cdot\|_\infty$.

$$\begin{bmatrix} 0.04 & 0.01 & -0.01 \\ 0.2 & 0.5 & -0.2 \\ 1 & 2 & 4 \end{bmatrix}$$

$K(A) = \|A\| \|A^{-1}\|$ ← definition

$\|A\|_\infty = \max\{0.04+0.01+0.01,\ 0.2+0.5+0.2,\ 1+2+4\}$
$= 7$

$A^{-1} = \frac{1}{\det A}\left[\text{Minors}\right]$

$\det A = 0.04\begin{vmatrix} 0.5 & -0.2 \\ 2 & 4 \end{vmatrix} - 0.01\begin{vmatrix} 0.2 & -0.2 \\ 1 & 4 \end{vmatrix} - 0.01\begin{vmatrix} 0.2 & 0.5 \\ 1 & 2 \end{vmatrix}$

$= 0.087$

$A^{-1} = \begin{bmatrix} 27.59 & -0.6897 & 0.03448 \\ -11.49 & 1.954 & 0.069 \\ -1.149 & -0.8045 & 0.2069 \end{bmatrix}$ $\|A^{-1}\|_\infty = 28.31418$

$K(A) = 7 \times 28.314 = 198.2$ ← ill conditioned.

Solve the following system with Gauss elimination with partial pivoting and scaling

$$\begin{aligned} x_1 - 5x_2 + x_3 &= 7, \\ 10x_1 + 20x_3 &= 6, \\ 5x_1 - x_3 &= 4. \end{aligned}$$

first scale

$$\left.\begin{aligned} \tfrac{1}{5}x_1 - x_2 + \tfrac{1}{5}x_3 &= \tfrac{7}{5} \\ \tfrac{1}{2}x_1 + x_3 &= \tfrac{6}{20} \\ x_1 - \tfrac{1}{5}x_3 &= 4/5 \end{aligned}\right\}$$

Then pivot

$$x_1 - \tfrac{1}{5}x_3 = 4/5$$

then pivot

$$x_1 - \frac{1}{5}x_3 = 4/5$$

$$\frac{1}{2}x_1 + x_3 = \frac{6}{20}$$

$$\frac{1}{5}x_1 - x_2 + \frac{1}{5}x_3 = 7/5$$

$$\begin{bmatrix} 1 & 0 & -1/5 & 4/5 \\ 1/2 & 0 & 1 & 6/20 \\ 1/5 & -1 & 1/5 & 7/5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1/5 & 4/5 \\ 0 & 0 & 1.1 & -0.1 \\ 0 & -1 & 0.24 & 1.24 \end{bmatrix}$$

pivoting $\Longrightarrow$ $\begin{bmatrix} 1 & 0 & -1/5 & 4/5 \\ 0 & -1 & 0.24 & 1.24 \\ 0 & 0 & 1.1 & -0.1 \end{bmatrix}$ No more scalling or pivoting is needed

$$x_3 = -\frac{0.1}{1.1} = -\frac{1}{11}$$

$$x_2 = -1.24 + \left(-\frac{1}{11} \times 0.24\right) = -1262$$

$$x_1 = 4/5 + \frac{1}{5}\left(\frac{1}{11}\right) = 0.7818$$